

C Programming

CSIT — 1st Semester — 2080 — Answer Sheet
Subject Code: CSC115

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define structure and nested structure. Write a program to find out whether the nth term of the Fibonacci series is a prime number or not. Read the value of n from the user and display the result in the main function. Use separate user-defined function to generate nth Fibonacci term and to check whether that number is prime or not.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain the relation to array and pointer. Differentiate between call by value and call by reference with a suitable program.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Differentiate between source code and object code. Create a structure named Book with members Book_Name, Price and Author_Name, then take input for 10 records of Book and print the name of authors having the price of book greater than 1000.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Describe the different types of I/O functions used in file handling with syntax.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write a program to read P×Q matrix of integers and find the largest integer of each row and display it.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write a program to calculate the factorial of a given number using recursion.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Write a program to check wheather the entered word is palindrome or not.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. List different types of operators and explain any three of them.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Trace the output#include#includevoid main(){int i=0, k;for (k=5; k>=0; k--){i = i+k;}printf ("%d\n", i);getch ();}

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Write a program to compute the sum of first 10 even numbers using function.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is dynamic memory allocation? Explain with a suitable program.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write a program to initialize an array of dimension 10 and sent the numbers within the array in ascending order.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.